

Myeloma and the Kidney

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Myeloma is an uncommon malignancy; it accounts for 1% of total malignancies and 10% of hematologic malignancies in the United States, with a stable age-adjusted incidence of 40 per 1 million. The incidence of myeloma is twice as high in African Americans as in Whites, and is higher for males than females. Myeloma is a disease of older people, with the median age at diagnosis exceeding 65 years.¹ Myeloma is characterized by a neoplastic proliferation of plasma cell clones producing monoclonal immunoglobulin, which in 95% of cases includes an excess of the light chain (LC) component, which can be nephrotoxic. Various causes and manifestations of acute kidney injury (AKI) are possible in myeloma (Table 68.1), but the major risk is myeloma cast nephropathy (MCN), which is a medical emergency that requires prompt diagnosis and intervention to avoid irreversible kidney failure. Other monoclonal B-cell–derived kidney disorders have been collectively described as monoclonal gammopathy of renal significance (MGRS); they differ from myeloma because they do not depend on expansion of the clonal mass for their nephrotoxicity and usually lack the additional features characteristic of myeloma, such as bone disease.²

ETIOLOGY AND PATHOGENESIS OF MYELOMA

Plasma cells derive from mature uncommitted B cells and after antigen stimulus undergo heavy-chain class switching from μ (immunoglobulin M [IgM]) expression to α , δ , ϵ , and γ . Whole immunoglobulin production requires the intracellular assembly of two heavy chains and two LCs, either kappa (κ) or lambda (λ), to derive whole IgG, IgA, IgD, and IgE. Normally, LCs are excreted in slight excess, with a κ/λ ratio of approximately 2:1. In myeloma a clone of cells secretes excessive quantities of a specific immunoglobulin and/or LCs (the paraprotein or M-protein). All myelomas (whole immunoglobulin or LC only) arise from a preclinical phase of monoclonal gammopathy (monoclonal gammopathy of undetermined significance [MGUS]) into a clinical or symptomatic phase as a consequence of somatic mutations. Mutations have important implications for prognosis and treatment and are incorporated into diagnostic and prognostic stratification tools.³ Myeloma remains an incurable disease because of multiple subclones, which engender resistance to chemotherapy and account for the natural history of remission and relapse over time. Both deregulation of cell cycling and impaired apoptosis account for their progressive and dysfunctional accumulation within the bone marrow and occasionally other organs. Plasma cells express little surface immunoglobulin and are recognized by surface expression of CD38 and CD138; they normally reside only in the bone marrow. In myeloma, unrestrained plasma cell growth is supported by a complex milieu of autocrine and paracrine cytokines, especially interleukin-6 (IL-6). These cytokines are secreted from stromal cells, endothelial cells,

and/or osteoclasts and maintain myeloma cell growth, survival, and migration; they also contribute to local organ dysfunction, such as bone resorption, fracture, and anemia.⁴

ETIOLOGY AND PATHOGENESIS OF KIDNEY DISEASE IN MYELOMA

Free light chains (FLCs) circulate as monomers (predominantly $\kappa \approx 25$ kDa) and dimers (predominantly $\lambda \approx 50$ kDa) with a very short half-life (2–6 hours) because of free glomerular filtration, whereas the much larger whole immunoglobulin circulates intact for several weeks. The filtered FLC is reabsorbed in the proximal tubule cell (PTC) by receptor-mediated endocytosis after binding with the glycoprotein receptor cubilin⁵ (Fig. 68.1). LCs in excess can induce an inhibitory effect on endocytosis in vitro and are associated with lysosomal overload and rupture, releasing enzymatic contents into the cytosol, manifested histologically by crystallization, vacuolation, and desquamation of the PTC. Endocytosis of LCs induces the release of the proinflammatory cytokines IL-6, IL-8, and monocyte chemoattractant protein-1 (MCP-1) by activation of nuclear factor- κ B (NF- κ B) in the PTC.⁶ Inhibition of proximal tubule uptake of LCs by silencing the megalin/cubilin genes with small interfering RNAs prevents proximal tubular toxicity.⁷ This suggests that LC overload induces factors promoting interstitial injury and fibrosis, as described in other proteinuric states. Less common manifestations of PTC injury include the Fanconi syndrome, which is invariably associated with specific variant κ LCs and often with pathologic evidence of crystalline inclusions.⁸

Injury to the PTC allows escape or overflow of LCs to the distal nephron, where they can interact with uromodulin (Tamm-Horsfall protein) secreted by the cells of the thick ascending loop of Henle. Variations in the specificity of the binding region of different LCs modify the affinity of the LC for binding with uromodulin, which could in part explain the variable nephrotoxicity of LCs.⁹ This specificity of the individual LC for uromodulin was illustrated by the finding that the intraperitoneal instillation of LCs isolated from humans with specific kidney LC-associated disease induces the same kidney injury in animals.¹⁰ Inhibiting the binding of LCs to uromodulin prevents cast formation.¹¹ Not all LCs are associated with tubular injury, and neither the amount nor type of urinary LCs correlates with the severity of cast formation. Nevertheless, in general, the higher the urinary excretion of LCs, the greater is the risk for kidney failure and reduced response to chemotherapy.^{12–14} In addition to LC-specific factors, tubular solute composition and tubular flow rates modulate the risk for cast formation. In animals, urinary acidification, furosemide, and urinary sodium and calcium concentrations may affect the tendency to increased binding or aggregation of LCs with uromodulin, whereas colchicine may reduce this tendency in animals but not in humans.^{15,16} The formation and passage of casts distally can occlude the tubule and

TABLE 68.1 Etiology of Kidney Injury and Clinical Manifestations in Myeloma

	Cause	Manifestation
Prerenal		
Volume depletion	Hypercalcemia	Polyuria and polydipsia
	Gastrointestinal losses (nausea and vomiting)	Hypotension
	Sepsis	Fever
Hemodynamic	Hemodynamic effects of NSAIDs	Oliguria, hyperkalemia
Other	Hyperviscosity (IgA, IgG3)	Mental state alterations
	Hyperuricemia	Tumor lysis
Intrarenal	Proximal tubular injury from light chains, uric acid, distal tubular injury from casts	Fanconi syndrome Tubular proteinuria Crystalluria
	Glomerular disease (LCDD, amyloid)	Nephrotic proteinuria Hematuria, active sediment
Postrenal	Calculi	Colic

Ig, Immunoglobulin; *LCDD*, light chain deposition disease; *NSAIDs*, nonsteroidal antiinflammatory drugs.

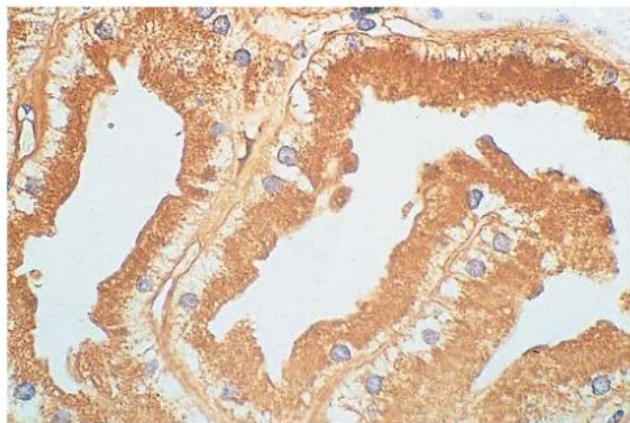


Fig. 68.1 Uptake of Light Chains by Proximal Tubular Cells. Kidney biopsy from a patient excreting κ light chains. Immunoperoxidase staining (brown color) showing κ light chains along the brush border and in the cytoplasm of proximal tubular cells.

allow intratubular obstruction, with rupture and even backflow of contents (Fig. 68.2).

EPIDEMIOLOGY

Most myeloma manifests *de novo*, although around 1% evolves from patients with a known MGUS each year. In patients with newly diagnosed myeloma, the prevalence of IgG, IgA, IgD, and FLC-only myeloma was 52%, 21%, 2%, and 16%, respectively.¹³ IgM and IgE myeloma are extremely uncommon. Approximately 70% of patients with myeloma also have a urinary M-protein. At the time of diagnosis, up to 50% of patients have evidence of impaired kidney function judged by increased serum creatinine, around 25% presenting with serum creatinine greater than 2 mg/dL (177 μ mol/L).^{13,14} In unselected series, 2% to 10% of patients present with severe kidney failure requiring dialysis; this figure is higher in series reported from nephrology units. In contrast to the general distribution of M-protein types in myeloma, LC and IgD myeloma are particularly associated with higher

risk for kidney disease and are present in nearly 50% of patients with kidney failure requiring dialysis.¹⁷

CLINICAL PRESENTATION

Most patients present with constitutional symptoms (fatigue, weight loss) and skeletal pain, especially back pain. Kidney impairment is common and has a variety of causes (see Fig. 68.1). In a smaller proportion of patients, kidney failure is the presenting manifestation of myeloma and these patients often have more advanced disease with high morbidity/mortality.¹⁷ Kidney findings are nonspecific, typically normal-size kidneys and bland urine. Urinary protein excretion may be increased because of the presence of LCs (Bence Jones protein). Urinary dipstick (or measurement of albuminuria) may indicate only small amounts of protein because they measure albumin only. Significantly increased levels of urinary albumin are suggestive of amyloidosis or other monoclonal kidney disorders, which affect the glomerulus (see Chapter 28). Normal or elevated ionized calcium, a decreased serum anion gap, lytic bone lesions on radiographs, hypogammaglobulinemia, reduced levels of other immunoglobulin classes (immunoparesis), significant cytopenias, or blood film changes (plasma cells and/or leukoerythroblastic film) are suggestive of myeloma. Clinical and laboratory findings that may distinguish MCN from other monoclonal B-cell disorders affecting the kidney are listed in Table 68.2.

PATHOLOGY

Histologic examination of the kidney in myeloma and AKI has diagnostic and prognostic utility, although it is not always required.¹⁸⁻²¹ The introduction of the serum (FLC) ratio as a rapid diagnostic tool has reduced the requirement for biopsy for diagnosis. Cast nephropathy is the most common histologic finding, occurring in 30% to 50% of patients; however, a diverse range of diseases have been reported depending on the indication for biopsy (Table 68.3). Cast nephropathy is characterized by many distal tubular casts, which are strongly eosinophilic and consist of the monoclonal LC and laminated uromodulin, which often appear fractured after fixation. Casts promote local inflammation with giant cell formation.^{22,23} In 30% of cases, cast formation may not be prominent despite extensive tubulointerstitial injury²⁴ (Box 68.1 and Fig. 68.3). Glomeruli are usually spared unless there is associated LC deposition disease or amyloidosis (see Chapter 28).

DIAGNOSIS AND DIFFERENTIAL DIAGNOSIS

The International Myeloma Working Group (2014) diagnostic criteria for symptomatic myeloma require one or more myeloma-defining events in addition to bone marrow clonally restricted plasma cells of 10% or greater or biopsy-proven plasmacytoma. A myeloma-defining event is a CRAB event (hypercalcemia, renal failure, anemia or lytic bone lesions) or one of three biomarkers: clonal bone marrow plasma cells greater than 60%, serum FLC ratio of 100 or greater (involved to uninvolved LC and the involved LC absolute value >100 mg/L), or more than one focal bone lesion on magnetic resonance imaging. To meet the criteria for kidney failure as a myeloma-defining event, a reduction in estimated glomerular filtration rate (eGFR) to less than 40 mL/min/1.73 m² and a histologic diagnosis of cast nephropathy (or presumptive diagnosis based on a serum involved FLC of typically >1500 mg/L) is necessary to distinguish this from other paraprotein-defined kidney diseases.²⁵ About 97% of patients with a diagnosis of myeloma have a detectable intact whole immunoglobulin and/or a free LC by serum protein electrophoresis (SEP) and immunofixation

Kidney Injury Caused by Light Chains

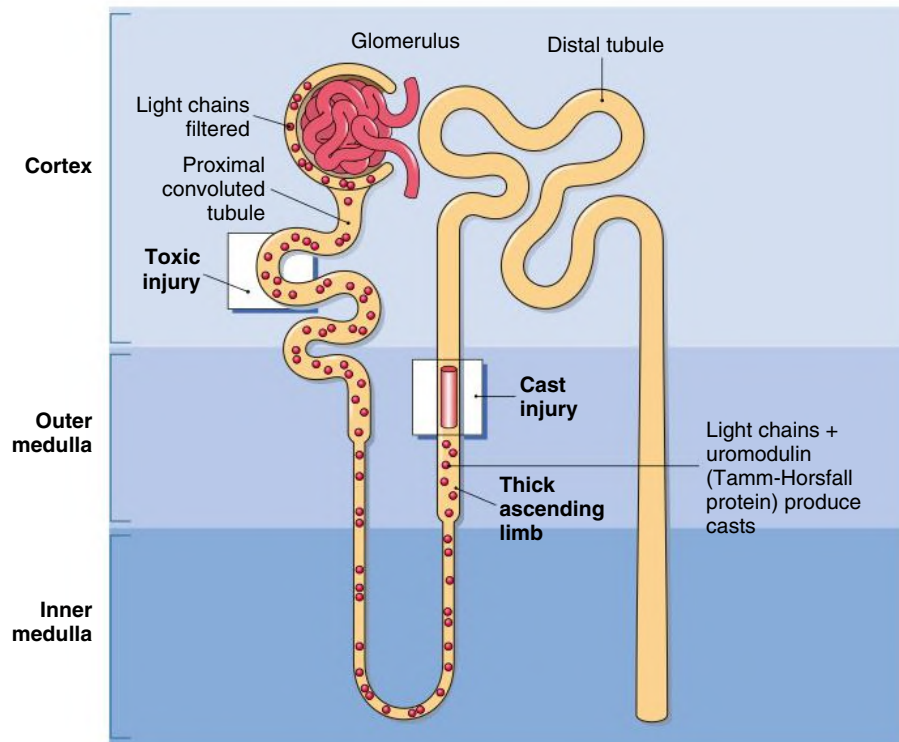


Fig. 68.2 Kidney Injury Caused by Light Chains. Sites (white boxes) where light chains injure the tubule. In the proximal tubule, there is direct tubular cytotoxicity. In the distal tubule, there is cast injury.

TABLE 68.2 Differentiating Features of Myeloma Kidney and Other Monoclonal Immune Deposition Diseases

	Myeloma Kidney	Other MIDDs
Proteinuria	<3 g/L	>3 g/L
Hematuria	Rare	LCDD, occasional Amyloidosis, rare
Hypercalcemia	Common	Absent
Hypertension	Uncommon	LCDD, common Amyloidosis, uncommon
Cytopenias	Anemia very common Leukopenia and thrombocytopenia, occasional	Uncommon
Immunoparesis ^a	Very common	Uncommon
Lytic bone lesions	Very common	Absent
Reduced GFR	Common	Common
Associated whole immunoglobulin	IgA, IgD, IgG	None
Type of light chain	Either	Amyloid $\lambda > \kappa$ LCDD $\kappa > \lambda$
Serum light chain elevation	>500 mg/L	<500 mg/L

^aImmunoparesis, reduction in noninvolved immunoglobulin classes.
Ig, Immunoglobulin; LCDD, light chain deposition disease; MIDD, monoclonal immunoglobulin deposition disease; GFR, glomerular filtration rate.

(IFE). The quantity of the M-protein is estimated from the SEP and may be used both for diagnosis and to monitor response to therapy. Urinary LC (Bence Jones protein) is determined from a concentrated sample, but despite this LC may be below the level of detection by IFE. The specific quantitative FLC assay (which measure only LC not bound to whole immunoglobulin in serum) is significantly more sensitive than SEP and IFE alone and is an automated assay that allows rapid diagnosis.^{26,27} An abnormal monoclonal FLC increases the specific LC fraction and may suppress the other LC, which alters the normal ratio of FLC (0.26–1.65 κ/λ) to reflect the oversecretion of the abnormal LC clone (<0.26 for λ FLC clone, >1.65 for κ clone). The normal serum levels of FLC are very low (7–13 mg/L), but in patients with reduced GFR, accumulation of both (κ and λ) FLCs occurs as a result of reduced excretion, and because λ free light chains tend to circulate as dimers, their clearance is reduced more than the monomeric κ FLC. The use of an extended reference range of 0.37 to 3.1 improves diagnostic sensitivity for myeloma to 99% with 100% specificity in the presence of acute kidney failure requiring dialysis.²⁸ In patients with kidney failure and biopsy-proven cast nephropathy, the ratios are highly abnormal, and the absolute increase in the affected LC is of the order of 100 to 200 times normal with values greater than 500 mg/L at presentation and a median of 7000 mg/L^{28,29} (Fig. 68.4). Measurement of serum FLC has significant advantages over SEP and IFE and is now incorporated into international clinical guidelines for the diagnosis and management of myeloma.³⁰

Diagnostic difficulties can arise when elderly patients who are evaluated for newly diagnosed kidney impairment have an M-protein on routine SEP. Approximately 3% of the population older than 70 years

TABLE 68.3 Kidney Pathology in Patients With Myeloma

Histologic Finding	Prevalence (%)
Myeloma kidney (myeloma cast nephropathy)	30–50
Interstitial nephritis/fibrosis without cast nephropathy	20–30
Amyloidosis	10
Light chain deposition disease	5
Acute tubular necrosis	10
Other (urate nephropathy, tubular crystals, hypercalcemia, focal segmental glomerulosclerosis)	5

Data from Nasr SH, Valeri AM, Sethi S, et al. Clinicopathologic correlations in multiple myeloma: a case series of 190 patients with kidney biopsies. *Am J Kidney Dis.* 2012;59(6):786–794; Herrera GA, Joseph L, Gu X, Hough A, Barlogie B. Renal pathologic spectrum in an autopsy series of patients with plasma cell dyscrasia. *Arch Pathol Lab Med.* 2004;128(8):875–879; and Sanders PW, Herrera GA, Kirk KA, Old CW, Galla JH. Spectrum of glomerular and tubulointerstitial renal lesions associated with monotypic immunoglobulin light chain deposition. *Lab Invest.* 1991;64(4):527–537.

BOX 68.1 Histologic Features of Myeloma Cast Nephropathy

Many eosinophilic, often fractured casts (medullary portion of the distal nephron predominantly)
 Intratubular and interstitial macrophages and giant cells in response to casts
 Interstitial inflammation, fibrosis, tubular atrophy, crystalline inclusions
 Minimal glomerular abnormality
 Minimal or no vascular changes

will have a serum M-protein, most consistent with MGUS. A diagnosis of MGUS is more likely if the serum level of the M-protein is low (<3 g/dL), there is absent or very low urinary LC (<1 g/24 h), and the absence of end-organ injury (no lytic lesions, <10% plasma cells on bone marrow aspirate).³¹ The FLC ratio should be normal using an extended reference range (0.82–3.6) for patients with eGFR less than 55 mL/min.³² This distinction is important because most patients with MGUS will die of an unrelated disease, and only 1% a year progress to myeloma.³³ Previously it was considered that most kidney disease in patients with MGUS was unrelated to the M-protein.³⁴ However, it is now recognized that the spectrum of paraprotein and LC disorders associated with specific kidney diseases histologically noted on biopsy is extensive and heterogeneous, but most are not associated with the malignant clonal expansion characteristic of myeloma. These disorders are now described as MGRS (see [Chapter 22](#)).² Evidence suggesting an alternative cause of the kidney disease (most commonly diabetes or vascular disease), along with a period of observation, may clarify the significance of the M-protein, although kidney and/or bone marrow biopsy is often required when diagnostic uncertainty persists.³⁵

NATURAL HISTORY

Most patients with reduced eGFR at presentation will show resolution of these predominantly functional changes with therapeutic measures that include withdrawal of nephrotoxins, rehydration, treatment of hypercalcemia, treatment of sepsis, and reduction in LC load with chemotherapy. Response to treatment with improvement in kidney function is associated with better clinical outcomes.³⁶ Reversibility occurs more frequently with higher GFR at presentation, lower LC

excretion, and presentation with hypercalcemia. The number of casts and degree of interstitial fibrosis/tubular atrophy present at biopsy are independently associated with kidney recovery, itself associated with improved patient survival.¹⁸ Although GFR improves in the majority of patients, approximately 10% of patients presenting with reduced GFR at diagnosis may require dialysis. Before the introduction of modern chemotherapy strategies, patients requiring dialysis had reported rates of recovery of kidney function as low as 5% to 15%. However, the introduction of specific LC reduction strategies has increased kidney recovery in up to 80% of patients if a significant reduction in the FLC level is achieved by day 21, with improvement in GFR and independence from dialysis being associated with improved patient outcomes.^{37,38}

TREATMENT

There are three key issues in the management of MCN. The first is to suspect and diagnose myeloma in the differential diagnosis of AKI. The inclusion of serum FLC measurement for routine evaluation of otherwise unexplained AKI in the appropriate clinical circumstances is essential. This is because kidney injury in myeloma is directly related to FLC excess and the increased sensitivity and specificity of FLC measurement, along with its rapid availability, may implicate or refute myeloma as a likely diagnostic possibility well in advance of results from SPE/IFE or kidney biopsy. Early diagnosis is crucial to allow implementation of the second strategy, which is to prevent or reverse oliguria by rapid identification and management of possible contributing factors to kidney impairment, which are present in around 50% of patients. Hypercalcemia, sepsis, and nonsteroidal antiinflammatory agents (NSAIDs) are the most common precipitants. Intravenous volume expansion is helpful to increase glomerular filtration, reduce single-nephron LC concentration, and increase tubular flow. The use of furosemide to promote diuresis should be avoided because it may favor cast formation. There is no clinical evidence of the superiority of volume expansion with sodium bicarbonate over sodium chloride, although prevention of urinary acidification is in theory desirable and in severe kidney failure may be necessary for management of metabolic acidosis. The maintenance of a high fluid intake (3 L/d) with water after initial volume correction and restoration of urine output with intravenous crystalloid is recommended to maintain high urine flow rates.

The third key requirement is the rapid reduction of FLC in the serum by chemotherapy. The immediate initiation of high-dose dexamethasone 40 mg/d is recommended, because plasma cells are highly responsive to corticosteroids, which induce rapid apoptosis and lowering of LC concentration. The addition of newer chemotherapy strategies (see later discussion) with high immediate LC-lowering effects in combination with corticosteroids is critical to achieving a rapid response.³⁰ Reducing the LC load with plasma exchange did not improve either recovery from kidney failure or patient survival.³⁹ The large volume of distribution of LC (which freely cross cell membranes) is not well suited to plasma exchange. The use of high cutoff dialysis membranes (molecular cutoff up to 50 kDa), in conjunction with chemotherapy, was initially shown to be associated with a reversal of kidney failure and the rapid reduction in circulating FLC in patients with cast nephropathy.^{38,40} However, two randomized controlled trials did not confirm an independent benefit of high cutoff membranes when combined with bortezomib chemotherapy, confirming that early response to chemotherapy is the more important factor.^{41,42} A suggested algorithm for management of myeloma with AKI is shown in [Fig. 68.5](#).

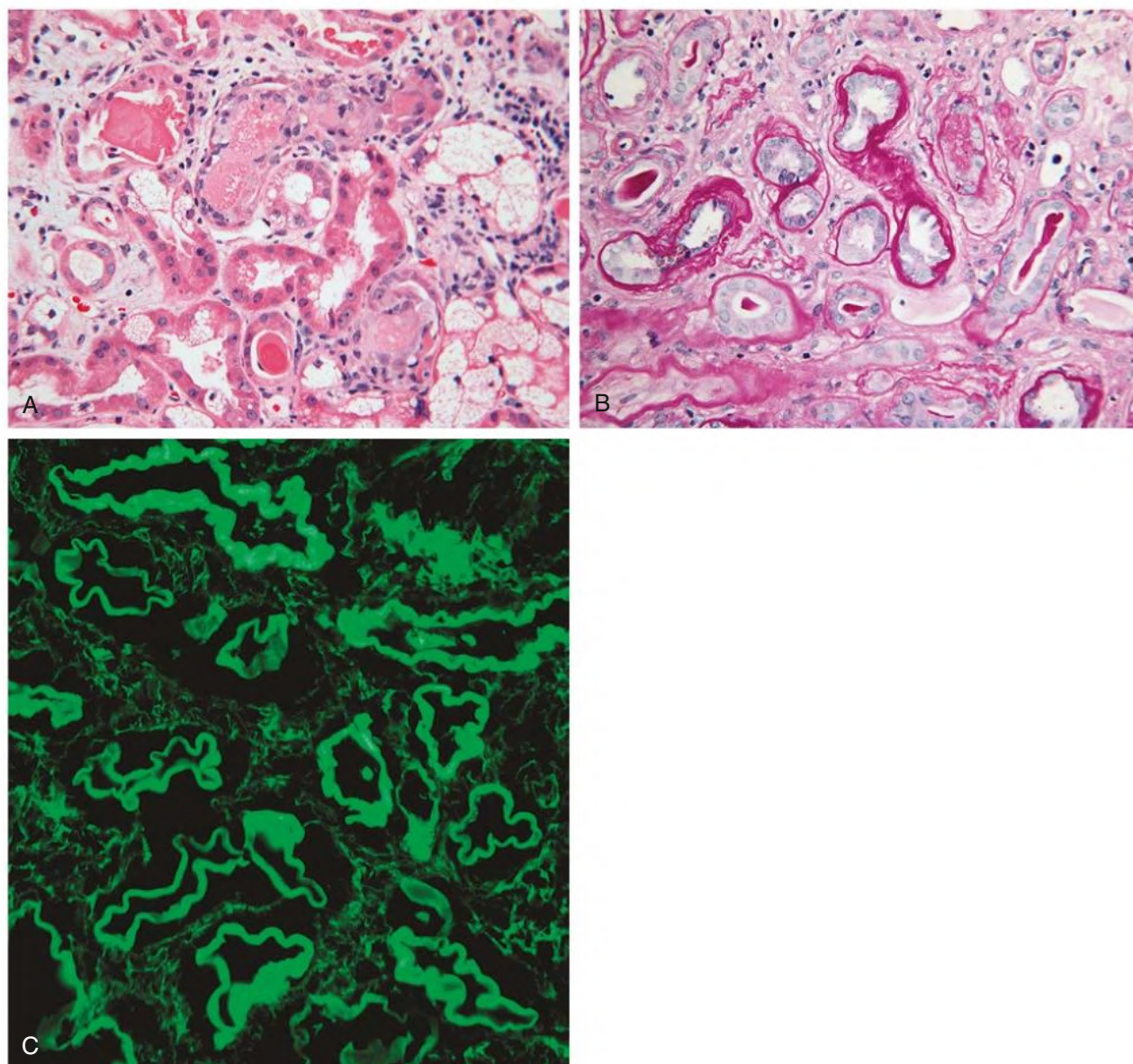


Fig. 68.3 Myeloma Cast Nephropathy. (A) Myeloma kidney. Many dilated tubules are obstructed by densely eosinophilic hard casts, with giant cell reaction and inflammatory cell infiltrates. There is also vacuolation and degeneration of tubules (hematoxylin and eosin stain; $\times 160$). (B) Light chain deposition along tubular basement membrane (TBM) in myeloma kidney without casts. There is marked thickening of the TBM by periodic acid-Schiff (PAS)-positive deposits and interstitial fibrosis and mild chronic inflammation (PAS stain; $\times 160$). (C) Light chain deposition along TBM. Strong linear deposition of κ light chain in the thickened TBM (direct immunofluorescence anti- κ light chain; $\times 160$). (Courtesy R. Sinniah, Perth, Australia.)

Chemotherapy

Although myeloma is incurable, chemotherapy will induce clinical improvement by control of the underlying disease in most patients. Chemotherapy involves an immediate induction treatment followed by consolidation strategy after disease control. Bortezomib, a reversible proteasome inhibitor whose clearance is independent of kidney function, has been successfully and safely used in patients with kidney failure and myeloma without significant toxicity and is the agent of choice for rapid reduction of LCs in combination with dexamethasone.³⁰ Bortezomib lowers the LC load by direct effects on plasma cells and also may protect kidney proximal tubular cells by inhibiting activation of IL-6 and NF- κ B induced by LC endocytosis.⁴³ Immunomodulatory drugs such as thalidomide, lenalidomide, and

pomalidomide are also frequently used for induction (triple therapy), as consolidation therapy, or for relapsed disease in conjunction with dexamethasone.³ Agents used for management of myeloma in patients with AKI are summarized in Table 68.4.⁴⁴ Autologous stem cell transplantation (ASCT) significantly extends survival and is now the treatment of choice for eligible patients.³⁰ Patients on dialysis are suitable for ASCT, which is associated with kidney recovery and freedom from dialysis, although with greater toxicity and increased morbidity and mortality.⁴⁵⁻⁴⁷ For patients unsuited for ASCT, chemotherapy selection depends on comorbidity, and dexamethasone/lenalidomide may be useful. Close multidisciplinary management and discussion with hematology and oncology staff is required to individualize decision-making because of the continuous advances in this area.

Serum Free Light Chain Ratio and Values

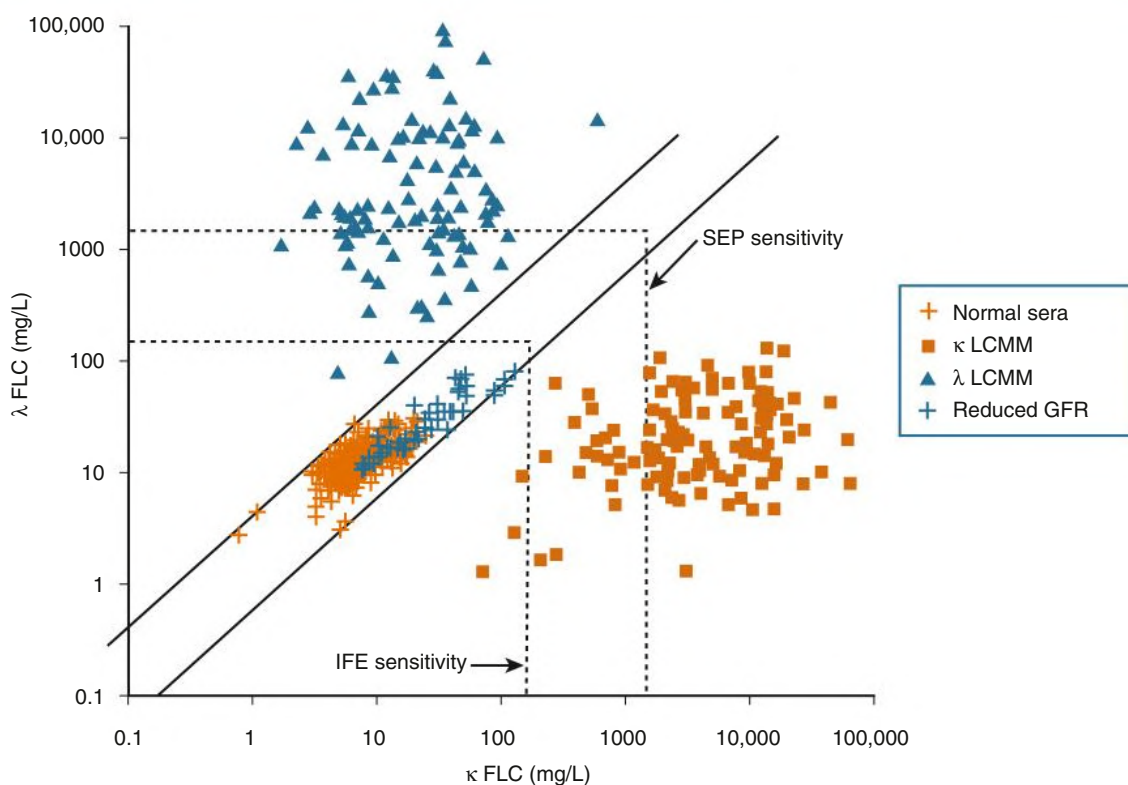


Fig. 68.4 Serum free light chain ratio and values in normal individuals, patients with reduced glomerular filtration rate (GFR) and light chain myeloma. *FLC*, free light chain; *IFE*, Immunofixation; *LCMM*, light chain multiple myeloma; *SEP*, serum protein electrophoresis.

Infection and progression of disease remain the major impediments to long-term survival in myeloma, especially within the higher-risk end-stage kidney disease (ESKD) group. Median survival in recent ESKD series with conventional chemotherapy before ASCT was only 4 to 8 months; however, survival of up to 7 years is reported in selected cases of ESKD managed with ASCT, and overall patient survival has improved with the advent of newer treatment strategies.^{48,49} The improved overall management of myeloma with prolonged survival of patients will increase the referral of patients with relapsed and treatment-refractory disease presenting with MCN. The management of this patient group is challenging and must be individualized based on overall prognosis and potential for salvage chemotherapy and likelihood of benefit from dialysis.

Adjunctive Therapies

Patients with myeloma and chronic kidney disease or kidney failure respond to erythropoiesis-stimulating agents (although may require higher doses), and these should be used along with other hematopoietic agents, including granulocyte colony-stimulating factor (G-CSF) as required, although early in the disease frequent transfusion may be necessary. Intravenous bisphosphonates rapidly correct hypercalcemia and are preferred to furosemide (which may aggravate tubular LC injury) and in the longer term reduce bone fracture.⁵⁰ However, there are several reports of toxicity (acute tubular necrosis, proteinuria) with full dose and/or rapid infusion of pamidronate and zoledronic acid; the use of reduced doses (pamidronate 30–60 mg, zoledronic acid 4 mg) and slower infusion rates is recommended.^{51,52} Alternatively,

bisphosphonates could be avoided in patients with eGFR less than 30 mL/min/1.73 m² by using denosumab to manage hypercalcemia.⁵³

DIALYSIS AND TRANSPLANTATION

Where kidney replacement therapy is required in the long term, both hemodialysis and peritoneal dialysis have been successfully used and there are no data to indicate that outcome is influenced by dialysis modality. Choice of long-term therapy requires individual assessment of patient circumstances. Aggressive measures to reduce the risk for systemic infection (antibiotic catheter locking solutions; mupirocin at exit sites) should be routinely considered because initially central venous catheters are often required for dialysis and chemotherapy. The timing of arteriovenous fistula placement requires an individualized risk assessment because it may be several months before irreversibility of kidney failure or a response to chemotherapy is proven. Kidney transplantation has a high risk for disease recurrence in the allograft and patient mortality, and given the incurability of the disease, has rarely been appropriate.⁵⁴ Current advances in chemotherapy and prolonged patient survival have prompted reconsideration of the role, timing, and management of kidney transplant.⁵⁵ The use of kidney transplant after successful induction therapy and ASCT may be considered for younger patients when disease remission by standard criteria and normalization of the serum FLC ratio is achieved. Allogeneic bone marrow and kidney allograft transplant have been reported, demonstrating the ability to induce allograft tolerance in haploidentical pairs.⁵⁶ This treatment is limited in availability and is not suitable for the majority of patients.

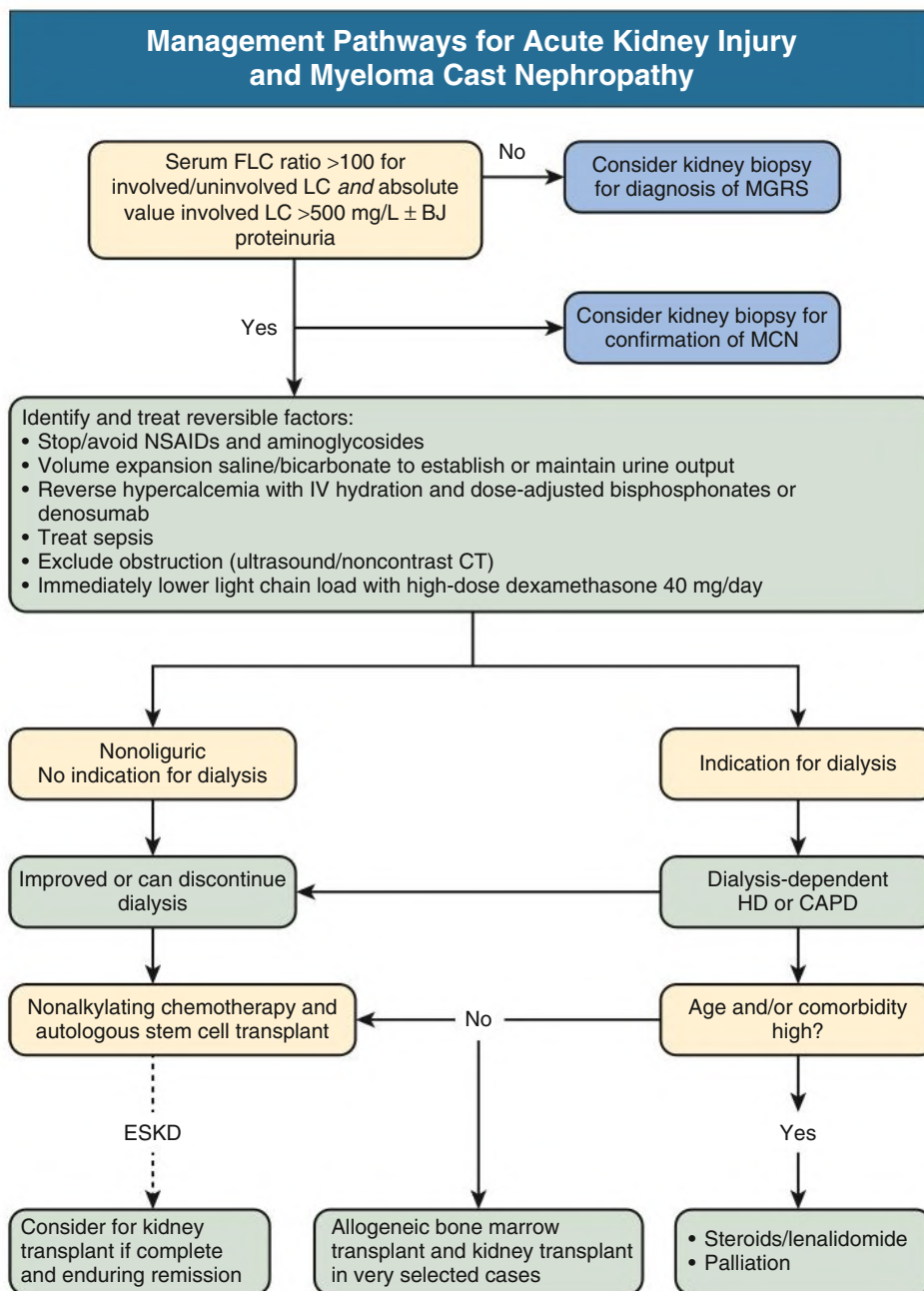


Fig. 68.5 Management Pathways for Acute Kidney Injury and Multiple Myeloma. *BJ*, Bence Jones; *CAPD*, continuous ambulatory peritoneal dialysis; *CT*, computed tomography; *ESKD*, end-stage kidney disease; *FLC*, free light chain; *HD*, hemodialysis; *IV*, intravenous; *LC*, light chain; *MCN*, myeloma cast nephropathy; *MGRS*, monoclonal gammopathy of renal significance; *NSAIDs*, nonsteroidal antiinflammatory drugs.

TABLE 68.4 Chemotherapeutic Drugs Used in Myeloma and Kidney Failure

Class	Agent	Route of Administration	Dose Reduction for CrCl <30 mL/min/1.73 m ²	Side Effects
Corticosteroids	Dexamethasone, Prednisone	PO/IV	None	Standard
Alkylating agents	Melphalan	PO/IV	Yes	Bone marrow suppression, infection
	Cyclophosphamide	PO/IV	No	
Immunomodulatory drugs	Thalidomide	PO	No	Neuropathy, venous thrombosis, hyperkalemia, cytopenias
	Lenalidomide	PO	Yes	
	Pomalidomide	PO	No	
Proteasome inhibitors	Bortezomib	IV/SC	None	Neuropathy, cytopenias, TMA
	Carfilzomib	IV	No	
	Ixazomib	PO	Yes	
Anti-CD38 monoclonal antibodies	Daratumumab	IV	No	Cytopenias
	Isatuximab			

CrCl, Creatinine clearance; IV, intravenously; PO, orally; SC, subcutaneously; TMA, thrombotic microangiopathy.

SELF-ASSESSMENT QUESTIONS

- Kidney injury in patients with myeloma is most usually due to an excess of which of the following?
 - Calcium
 - Uric acid
 - Light chains
 - Immunoglobulins
 - β_2 -Microglobulin
- What is the most common histologic lesion noted in patients with acute kidney injury and myeloma?
 - Amyloidosis
 - Light chain deposition disease
 - Cast nephropathy
 - Urate nephropathy
 - Tubular crystals
- In addition to corticosteroids, which therapy should be commenced urgently for the management of myeloma cast nephropathy?
 - Vincristine
 - Adriamycin
 - Melphalan
 - Cyclophosphamide
 - Bortezomib
- A 72-year-old man has a 3-month history of back pain and is managed with nonsteroidal antiinflammatory agents. He presents with anemia, hypercalcemia, and acute kidney injury. You suspect myeloma. What is the most appropriate investigation to rapidly confirm the diagnosis?
 - Urine Bence Jones protein
 - Serum protein electrophoresis
 - Serum immunoglobulins
 - Serum free light chains
 - Kidney biopsy

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